

Furkan Akin

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EDUCATION

University of Illinois at Urbana-Champaign <i>Bachelor of Science in Computer Science; GPA: 3.95 (In Progress)</i>	Champaign, IL August 2024 – May 2026
William Rainey Harper College <i>Associate of Science in Engineering; GPA: 4.0 (Final)</i>	Palatine, IL August 2022 – May 2024
Seattle Central College <i>Associate in Business; GPA: 3.8 (Final)</i>	Seattle, WA August 2021 – May 2023
Relevant Coursework: Database Systems - Distributed Systems - Python Programming for Data Science - Applied Machine Learning - System Programming - Artificial Intelligence - Probability and Statistics for CS - Computer Architecture - Data Structures and Algorithms	
Honors & Awards: Grainger Engineering James Honors Scholar - Dean's List - Tau Beta Pi Engineering Honors Society - John R. Pasta Outstanding Undergraduate Award - Engineering Pathways Scholarship	

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, NoSQL, JavaScript, TypeScript, HTML/CSS, Go
Frameworks\Developer Tools: Git, VS Code, IntelliJ, Flask, PyCharm, React, Android Studio, Node.js, Django, Spring Boot, Jira, Tableau, Docker, Terraform
Databases\Cloud Platforms: Google Cloud Platform, AWS, DBeaver, MySQL, PostgreSQL, SQLite, MongoDB, Neo4j
Libraries: pandas, NumPy, sci-kit, Matplotlib, PyTorch, TensorFlow, D3.js

EXPERIENCE

Software Engineering Intern <i>CNA Insurance</i>	May 2025 – August 2025 Chicago, IL
- Engineered 5 full-stack features for CNA's largest account platform, supporting \$2B+ in profit transactions. - Produced 66 REST APIs, 28 normalized DB tables, & 8 full-stack screens to streamline underwriting workflows. - Devised refactoring strategy for 25+ React files, cutting onboarding time by 40% through separation of concerns. - Collaborated with software engineers and business analysts in an Agile Scrum environment, driving a 24% increase in team productivity.	
Database Systems Course Assistant <i>Siebel School of Computing and Data Science</i>	January 2026 – Current Champaign, IL
- Improved student understanding of relational and NoSQL databases by leading weekly labs and office hours on SQL, MySQL, MongoDB, and Neo4j for 300 students, contributing to a 12% increase in average lab or exam scores. - Identified and resolved database performance issues by guiding students on indexing strategies and query optimization, reducing common query runtime issues by 25% across assignments.	
AI & Data Ethics Course Assistant <i>Siebel School of Computing and Data Science</i>	January 2025 – Current Champaign, IL
- Guided students on case studies involving AI misuse, algorithmic bias, and professional responsibility, supporting 200+ students in developing ethical reasoning skills. - Hosted weekly office hours and provided personalized feedback, leading to 20% average improvement in exam performance.	

PROJECTS

OzAI: Voice-Based Conversational AI Assistant <i>Python, OpenAI, Whisper</i>	January 2025
- Designed and implemented a modular voice assistant supporting speech-to-text, LLM-based reasoning, and text-to-speech, achieving end-to-end voice response latency under 2-3 seconds on local hardware. - Integrated OpenAI Whisper (STT) and cloud-based TTS (gpt-4o-mini-tts), enabling hands-free conversational interaction with 95% transcription accuracy for clear English speech in local testing. - Engineered a conversation memory system with context trimming, supporting 20+ turn multi-turn dialogue while preventing context-window overflow.	
RainStorm Distributed Stream Processing Framework <i>Go</i>	December 2025
- Built a fault-tolerant distributed stream processing system in Go, supporting multi-stage DAG execution across 10 worker VMs, processing tens of thousands of tuples per run with deterministic output. - Designed HyDFS, a custom distributed file system, to support exactly-once stream processing via durable tuple logging and selective replay, achieving 0 duplicate outputs under failures. - Developed a gossip-based membership and failure detection system with suspicion, detecting failures in 300–800 ms, enabling fast task reassignment and future autoscaling support.	
SpamBuster: GPT-Based Email Spam Classifier <i>Flask, OpenAI, scikit, Bash, Python</i>	December 2025
- Built an end-to-end email spam detection system using OpenAI GPT embeddings (text-embedding-3-small) and logistic regression, achieving 95% accuracy and 92% F1 score on the Enron dataset of 35,000 emails. - Automated environment setup and ML pipeline execution with bash scripts and virtual environments, reducing manual setup time from 30 minutes to under 2 minutes and ensuring reproducibility across multiple systems.	

CERTIFICATIONS

AWS Certified Cloud Practitioner <i>Amazon Web Services(AWS)</i>	January 2025 Credential Link
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